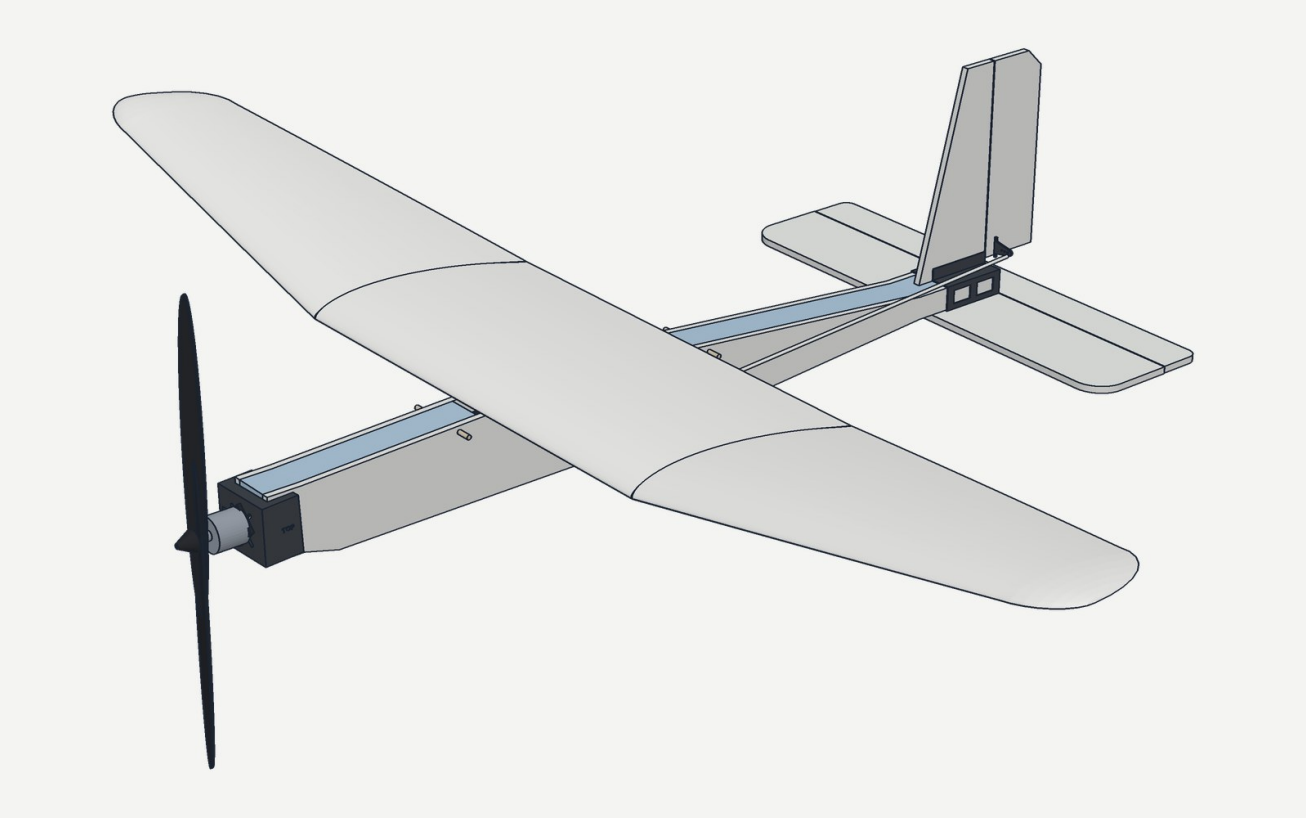
MERIDRON · BLUEPRINT CATALOG

D1b TRAINER

BUILD GUIDE · REV 1

25 Oct 2026 · Design checked · first flight pending



1160 mm
Wingspan

510 g
Take-off weight
with 3S 1300

≈ 27 min
Flight time
calculated, practice value

34 km/h
Cruise
calculated trim speed

≈ 16 h
Print time
rule of thumb 6 g/h

≈ 14 h
Hands-on build

≈ 233 CHF
Material
without electronics

EVERY MERIDRON PACKAGE WORKS THE SAME

1 **WATCH** the explainer video once

2 **DOWNLOAD** one ZIP, seven folders

3 **BUILD** step by step, each with a check

4 **CHECK** the list before the first flight

Calculated, not yet flown. Rev 1 becomes 'Flight proven' after the first build in the demonstrator workshop. Report anything that does not fit: every report makes the next revision.

BEFORE YOU START

TOOLS

- 3D printer (PETG, bed ≥ 200 × 200 mm)
- Sharp knife (snap-off blades) + steel ruler
- Sanding block (120 grit)
- Drill 4 mm
- Soldering iron (only if your ESC has no plugs)
- LiPo charger + LiPo bag
- Electric scroll saw with fine blades
- Flat building board ≥ 600 × 300 mm
- Covering iron (low temperature)
- Small screwdriver set, pliers
- Kitchen scale (1 g)
- Transmitter with ≥ 3 channels

SAFETY

• Charge LiPo batteries only in a LiPo bag, never unattended. Keep a fire extinguisher ready.

• Propeller off for every test on the bench. Battery in last, battery out first.

• Scroll saw and knife: cut away from your hand. Vacuum the foam dust.

• Epoxy and CA: gloves, open window. PETG instead of PLA: a cold garage makes PLA brittle.

WHAT TO BUY (CHF, ROUGH PRICES)

Group	Item	Qty	CHF
Foam	XPS sheet 30 mm (fuselage core)	1 sheet	15
Foam	Depron 6 mm (side walls, tail)	2 sheets (1 spare)	24
Foam	Depron 3 mm (wing skin)	1 sheet	14
Carbon	CFK tube 8/6 mm, 1 m (cut 380 mm)	1	12
Carbon	CFK rod 4 mm, 1 m (cut 2 × 294 mm)	1	5
Carbon	CFK rod 2 mm, 1 m (cut 390 + 2 × 332 mm)	2	6
Covering	Oracover Oralight film	1 roll	28
Glue	Foam-safe adhesive (wing skins)	1	7
Glue	UHU Por + 5-minute epoxy	1 + 1	19
Glue	Epoxy (joiners, nose)	1	9
Glue	Thin CA (superglue)	1	7
Print	PETG filament (uses ≈ 90 g)	1 kg	25
Tools	Fine scroll-saw blades	1 pack	10
Mechanics	2 pushrods in bowden tube + 2 connectors	1 set	12
Mechanics	Propeller 10 × 4.5	4	15
Mechanics	Fabric tape, velcro, rubber bands, bamboo skewers	1 set	15
Mechanics	M3 screws + nuts, XT60, 3.5 mm bullets	1 set	10
Electronics	Motor 2212 ~1000 kV with X-mount	1	—
Electronics	ESC 30 A with BEC	1	—
Electronics	Servo 9 g	2	—
Electronics	Receiver ≥ 4 channels (matching your transmitter)	1	—
Electronics	LiPo 3S 1300 mAh (2200 mAh also fits)	2	—
	Material total (without electronics)		233

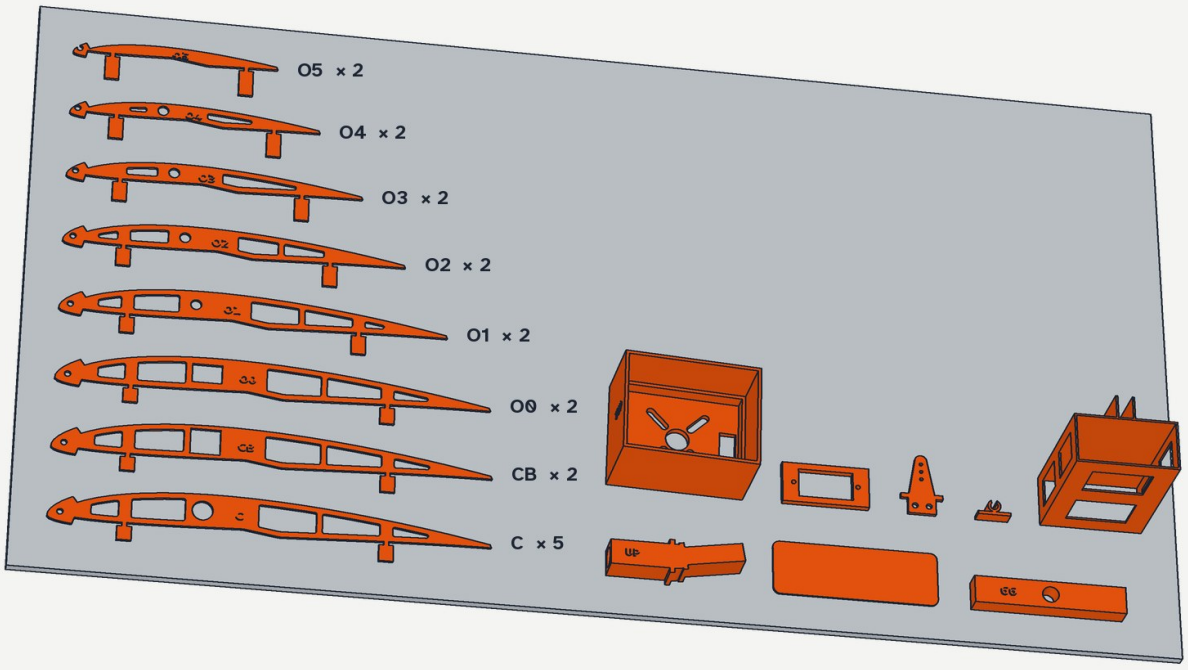
Prices: Oralight and foam glue read from a dealer on 23 Oct 2026, the rest estimated. Electronics are not priced: they depend on your radio system. Not included: shipping, customs, transmitter, charger.

01

STEP 1 OF 14 · PRINT

PRINT ALL PARTS

≈ 30 min



YOU NEED 02_Print (16 files) · PETG · 3D printer

DO

1. Open the print list. Print every file as many times as its name says (..._print-5x = 5 pieces).
2. PETG, 0.2 mm layers, 3 walls, 15 % gyroid infill. Ribs and plates lie flat as they come.
3. Do not scale. Switch off 'fit to bed'.
4. Clean the parts: break off the brim, keep the jig tabs on the ribs.

CHECK ✓

You have 35 parts plus the gauge. Every rib label (C, CB, O0–O5) is readable.

WATCH OUT

Scaling the files. One wrong rib size and the wing will not close.

≈ 16 h printer time (rule of thumb). You can cut foam while it prints.

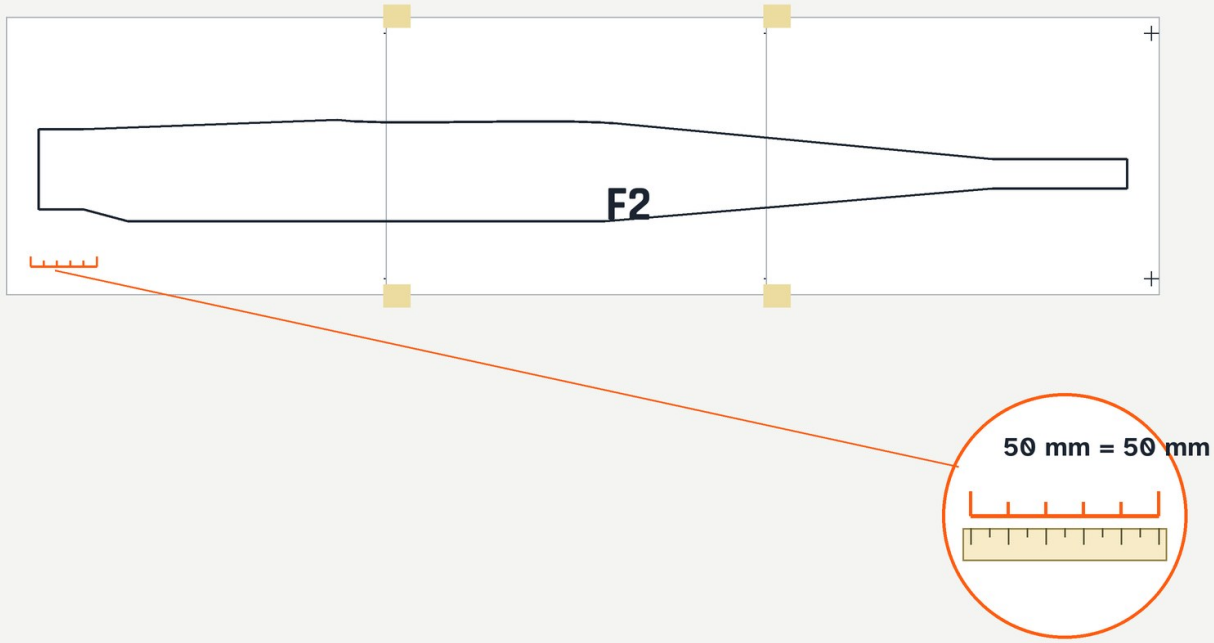
Step done ☐

PRINT AND JOIN THE TEMPLATES

≈ 40 min

Print at 100 %

grey band = overlap · line up the crosses · tape



YOU NEED 01_Cutting/...Templates_1to1_A4.pdf · Printer (A4) · Tape, scissors

DO

1. Print the template PDF at 100 % (actual size).
2. Measure the 50 mm bar on every page.
3. Tape the pages of each part together: line up the crosses in the grey overlap.
4. Cut the paper templates out on the thick line.

CHECK ✓

Every 50 mm bar measures 50 mm (± 0.5 mm). The 100 mm square on page 1 is square.

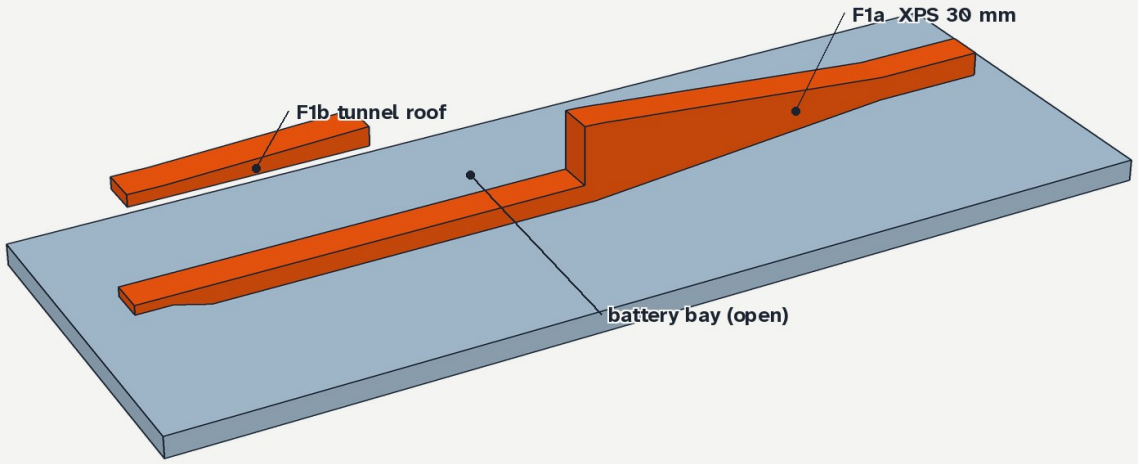
WATCH OUT

Printing with 'fit to page'. Everything ends up about 5 % too small.

Step done ☐

03 SAW THE FUSELAGE CORE

≈ 45 min



YOU NEED Template F1 · XPS 30 mm · Scroll saw, fine blade, vacuum

DO

1. Pin template F1 on the XPS sheet and trace it with a fine pen.
2. Saw 1 mm outside the line. Slow feed, let the blade cut.
3. Saw out the battery bay (the large opening under the wing).
4. Sand to the line with a flat sanding block.

CHECK ✓

The core lies flat on the table. Lay F1 on top: no gap larger than 1 mm.

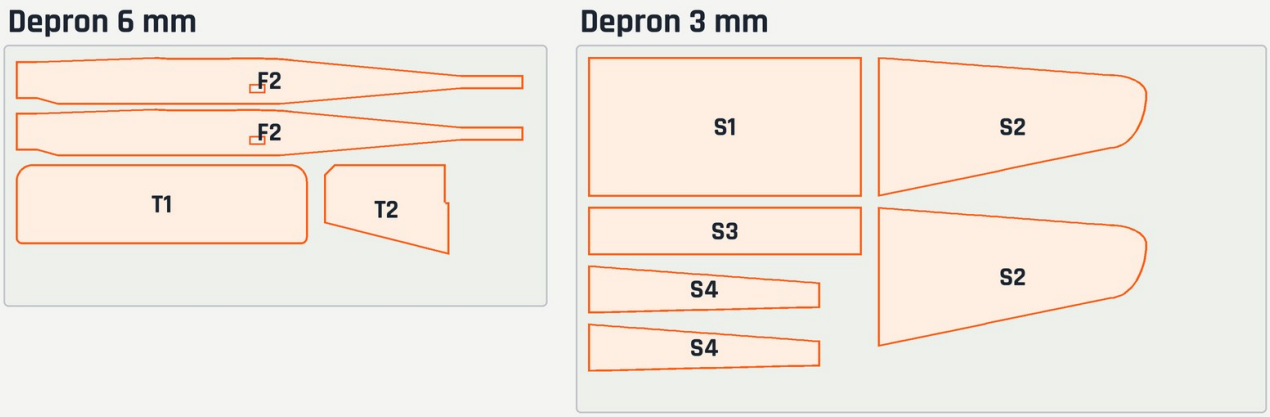
WATCH OUT

Pushing hard. The blade wanders and the edges become ragged.

Step done ☐

04 CUT THE DEPRON PARTS

≈ 60 min



Sharp blade, steel ruler, one clean cut. Orange holes = cut out.
Cut each S2 and S4 twice and flip one for the left wing.

YOU NEED Templates F2, T1, T2, S1–S4 · Depron 6 mm and 3 mm · Knife, steel ruler

DO

1. 6 mm: 2 × F2 side wall, 1 × T1 stabilizer, 1 × T2 fin.
2. 3 mm: 1 × S1 + 2 × S2 top skins, 1 × S3 + 2 × S4 bottom skins.
3. Cut the servo window in both side walls.
4. Mark the battery lines and dowel holes from the template on F2.

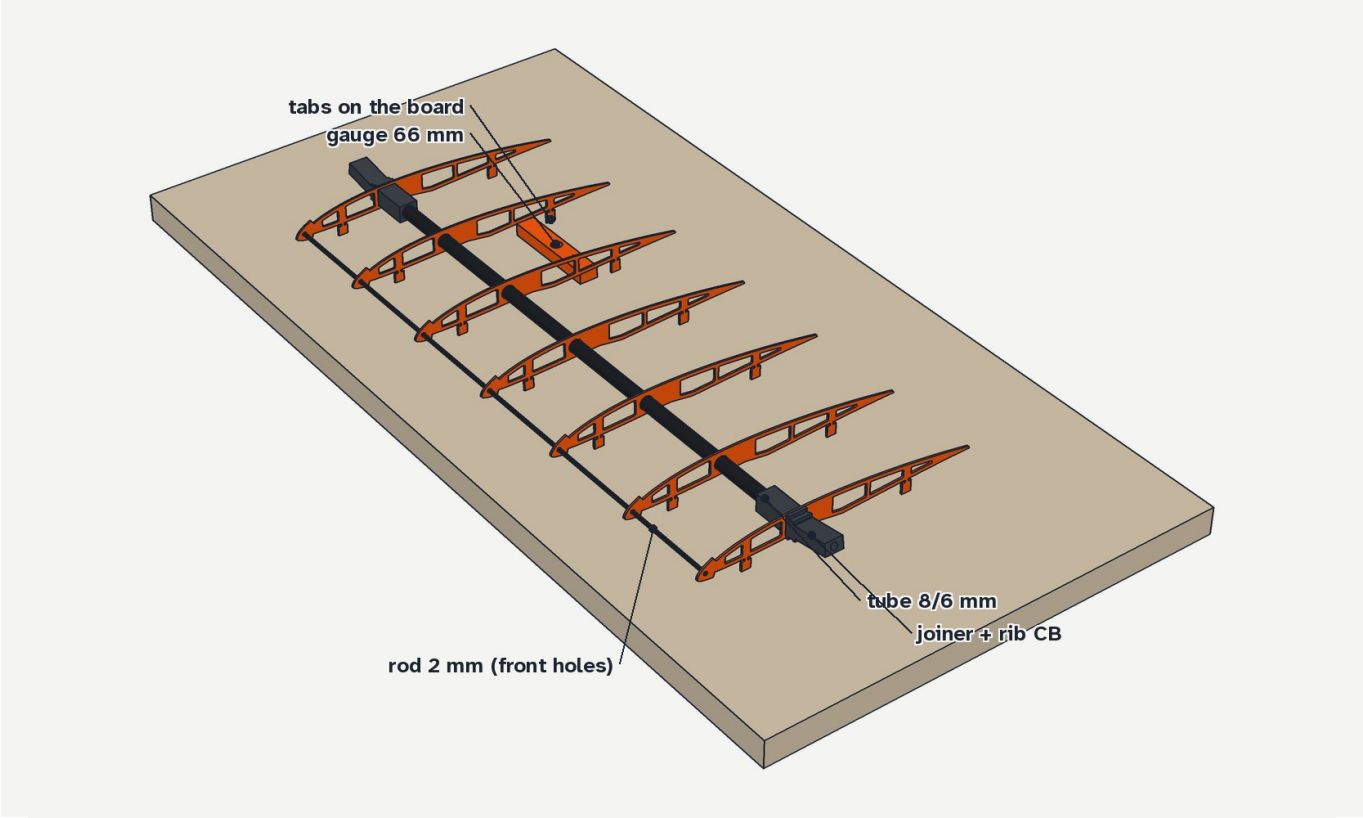
CHECK ✓
Lay each F2 on the core: outline matches within 1 mm, servo window above the bay.

WATCH OUT
A blunt blade tears Depron. Snap off a new tip every 2 m of cutting.

Step done ☐

BUILD THE CENTRE SECTION

≈ 60 min



YOU NEED 5 × rib C, 2 × rib CB · 2 joiners · Tube 8/6 (380 mm) · Rod 2 mm (390 mm) · Gauge, thin CA

DO

1. Push the 5 C ribs onto the tube. Push one joiner into each tube end; slide a CB rib onto each joiner up to the collar.
2. Stand everything on the board: all ribs rest on their two tabs.
3. Space the ribs with the 66 mm gauge, rib C in the middle.
4. Push the 2 mm rod through all front holes. Glue every joint with a drop of thin CA.

CHECK ✓

Every tab touches the board. Tube ends are equal on both sides (± 1 mm).

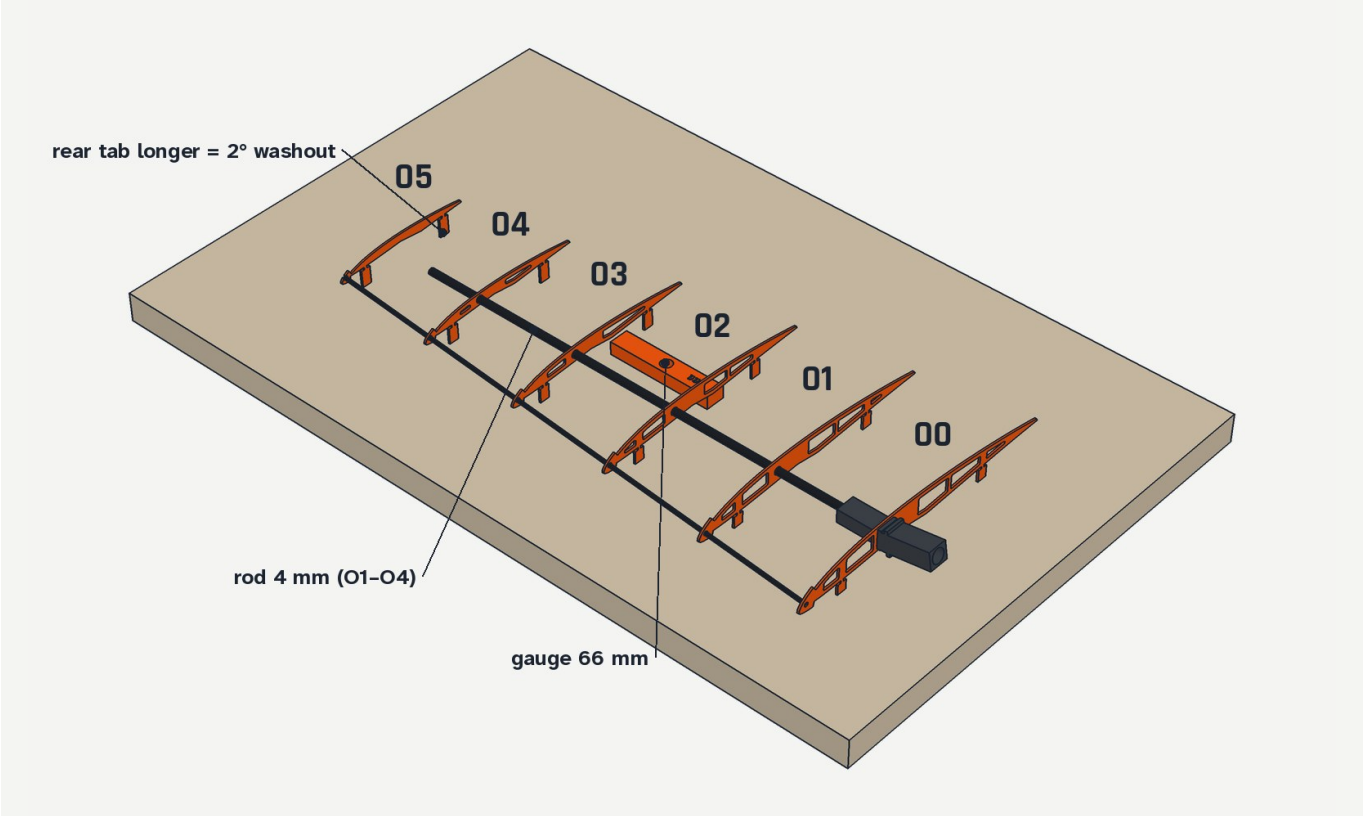
WATCH OUT

Building on a warped table. The board must be flat: the tabs copy it into the wing.

Step done ☐

06 BUILD THE TWO OUTER PANELS

≈ 1.5 h



YOU NEED 2 × ribs O0–O5 · Rod 4 mm (2 × 294 mm) · Rod 2 mm (2 × 332 mm) · Gauge, thin CA

DO

1. Lay the ribs out in order O0, O1 … O5. The label is on every rib.
2. Push the 4 mm rod through O1–O4. O0 and O5 stay without rod.
3. Stand the panel on its tabs, space with the gauge, push the 2 mm rod through all front holes.
4. Glue with thin CA. The tabs build the 2° washout in for you.

CHECK ✓

O5 (smallest rib): both tabs touch the board. The 4 mm rod ends about 30 mm before O5.

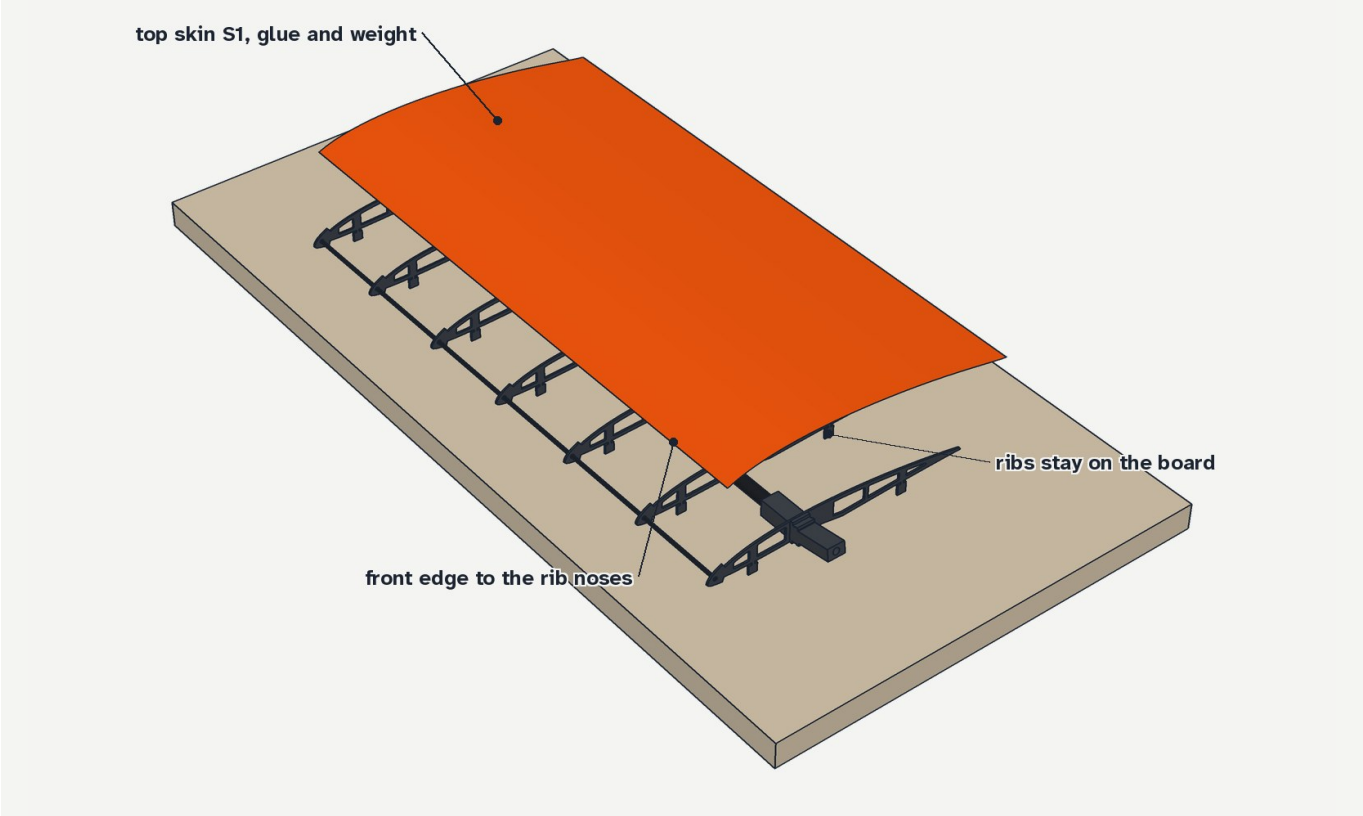
WATCH OUT

Mixing up the rib order. Follow the numbers, smallest rib at the tip.

Step done ☐

SKIN THE WING

≈ 2.0 h



YOU NEED Skins S1–S4 (Depron 3 mm) · Foam-safe glue · Books as weights

DO

1. Leave the ribs on the board. Glue the top skins S1 and S2 onto the ribs, front edge against the rib noses.
2. Weight evenly and let it dry fully.
3. Turn the panels over. Snap off all tabs and sand flush.
4. Glue the bottom skins S3 and S4 (front part only, up to the marked line).

CHECK ✓

The skin front edge touches every rib nose along the whole span. No gap over 1 mm.

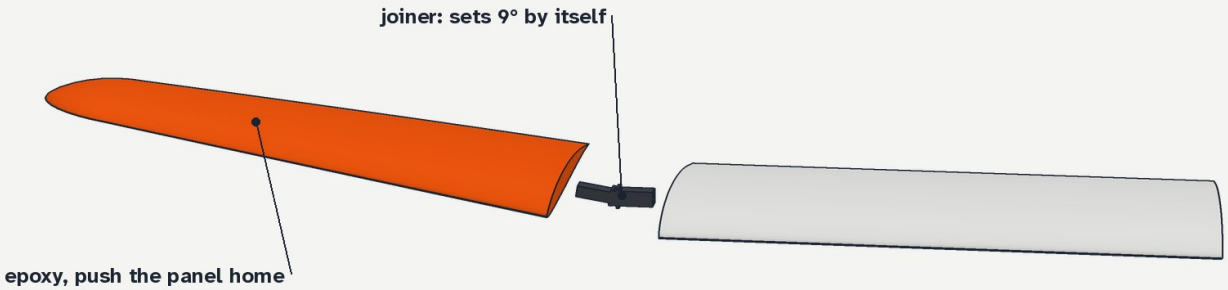
WATCH OUT

Too much glue. A thick layer warps Depron. Use a thin, even coat.

Step done ☐

JOIN THE PANELS

≈ 30 min



YOU NEED 5-minute epoxy · Centre section, 2 outer panels

DO

1. Put epoxy on the outer leg of each joiner and inside the OO rib window.
2. Push each outer panel onto its joiner until OO touches the collar.
3. Lay the centre flat on the table. The joiner sets 9° by itself.
4. Sand the top skins at the break until they meet.

CHECK ✓

Both tips stand 52 ± 5 mm above the table at the closing rib.

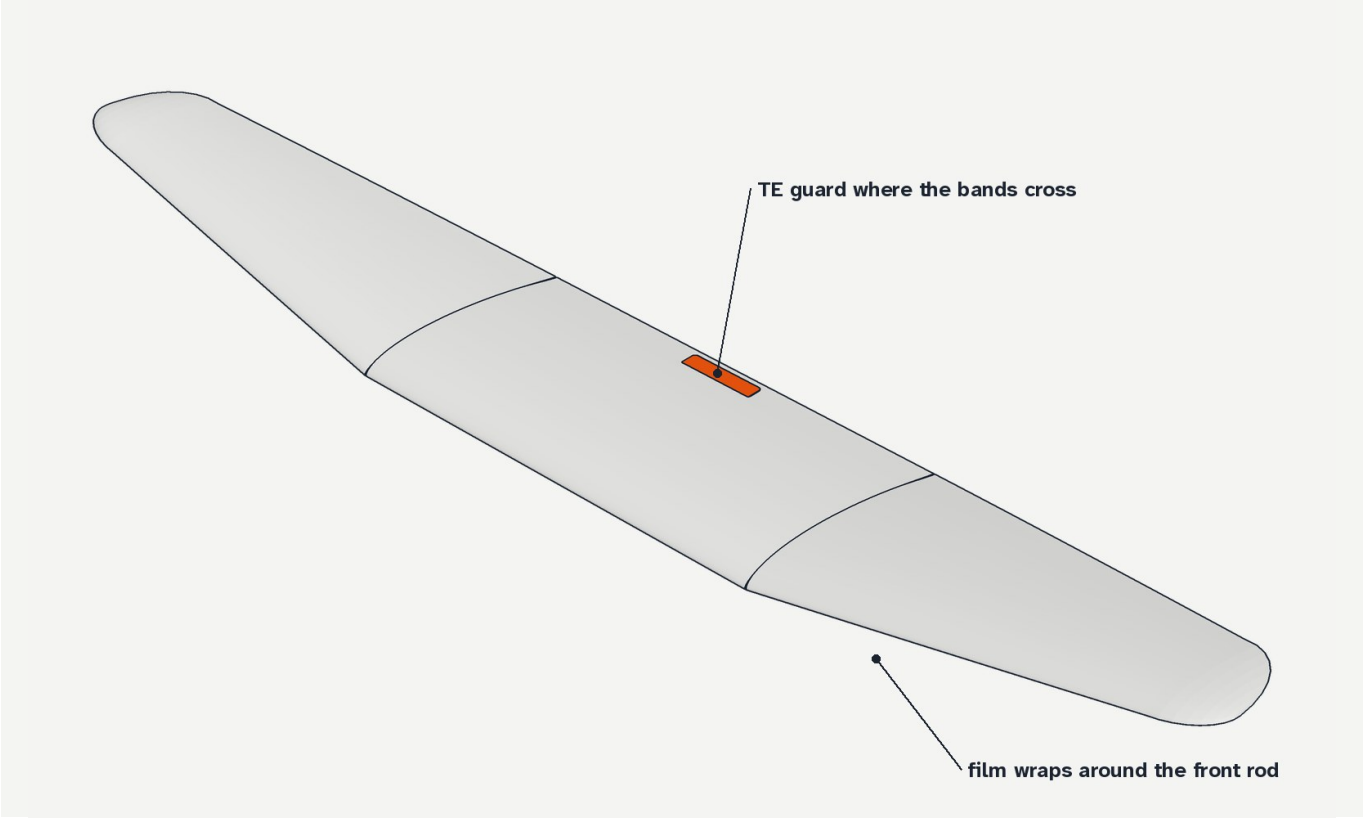
WATCH OUT

Epoxy on the outside of the skin. Wipe it off before it cures.

Step done ☐

COVER THE WING

≈ 1.5 h



YOU NEED Oralight film · Covering iron · Scrap pieces (test) · TE guard

DO

1. Test first on scrap: Depron alone, Depron on a rib, film over a printed part (tests T1–T3).
2. Start with the iron at its lowest setting. Raise only if the film does not stick.
3. Cover the bottom, then the top. Wrap the film around the 2 mm front rod.
4. Glue the TE guard on top at the centre where the rubber bands cross.

CHECK ✓

The test pieces show no dents. The wing lies straight; only the tips twist (washout).

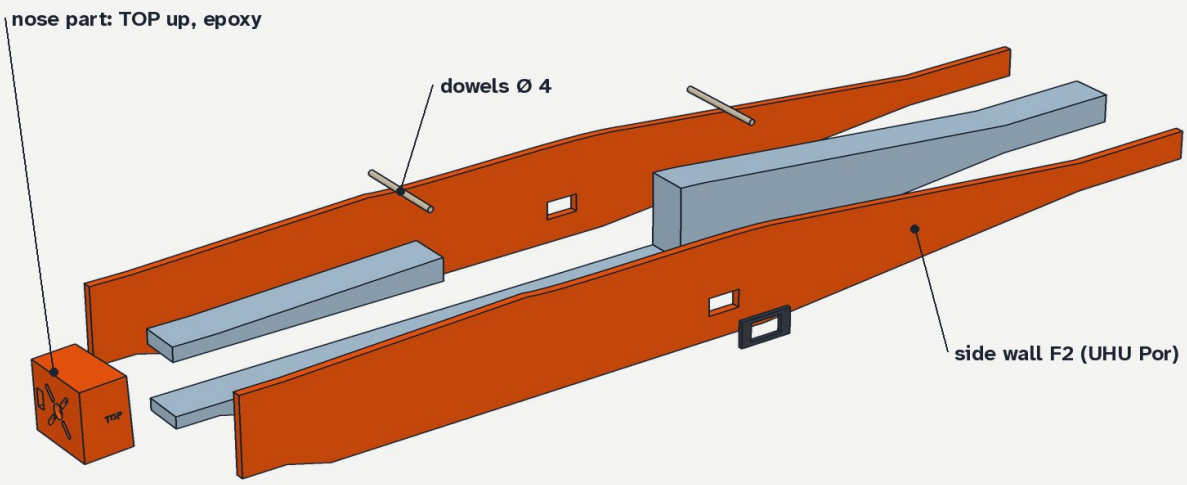
WATCH OUT

Too much heat. Depron collapses and the skin gets dents that stay.

Step done ☐

ASSEMBLE THE FUSELAGE

≈ 60 min



YOU NEED Core F1, 2 × side wall F2 · Nose part, 2 servo frames · UHU Por, epoxy · Bamboo skewers, drill 4 mm

DO

1. Coat core and walls thinly with UHU Por. Wait 10 min, then press together once.
2. Glue the servo frames on the outside around the servo windows.
3. Push the nose part on (word TOP up) and glue it with epoxy.
4. Drill the two dowel holes at the marks. Push the skewers through: 10 mm out on each side.

CHECK ✓

Sight from the nose: the fuselage is straight. The nose part sits flush all around.

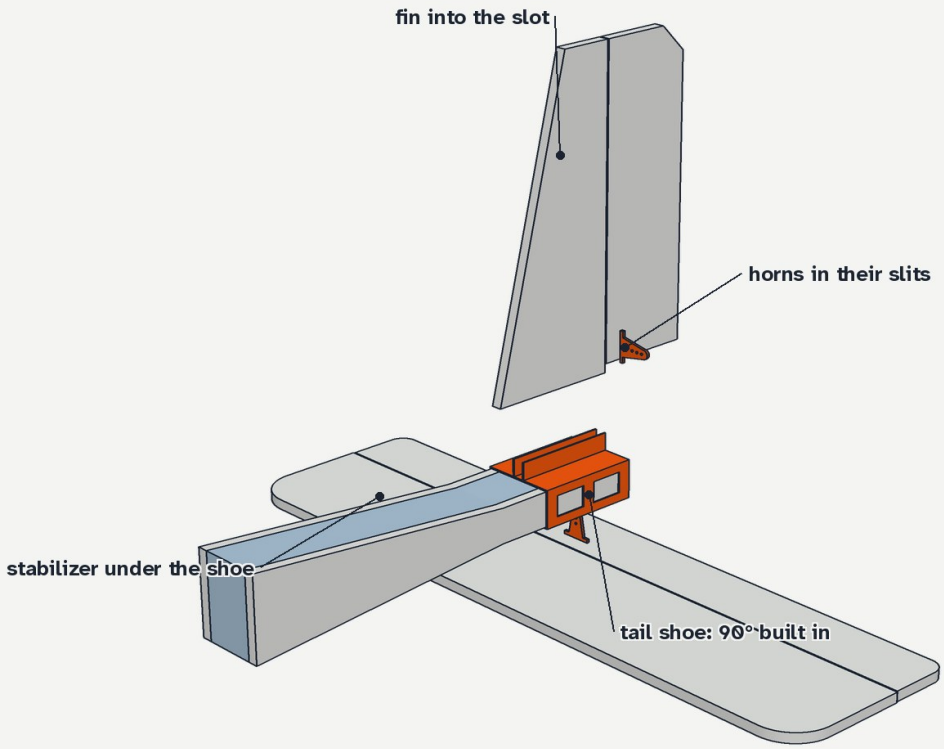
WATCH OUT

Pressing UHU Por wet. Contact glue only holds after it flashes off.

Step done ☐

FIT THE TAIL

≈ 60 min



YOU NEED T1 stabilizer, T2 fin · Tail shoe · 2 horns · Hinge tape, epoxy

DO

1. Cut T1 and T2 at the hinge line. Bevel 45°, hinge with tape.
2. Push the tail shoe onto the fuselage end, flush with the end. Glue.
3. Glue the stabilizer under the shoe and the fin into the slot on top.
4. Glue the horns into their slits: elevator right side, rudder left side.

CHECK ✓

Fin stands at 90° to the stabilizer (the shoe sets it). Seen from behind, stabilizer parallel to the wing seat.

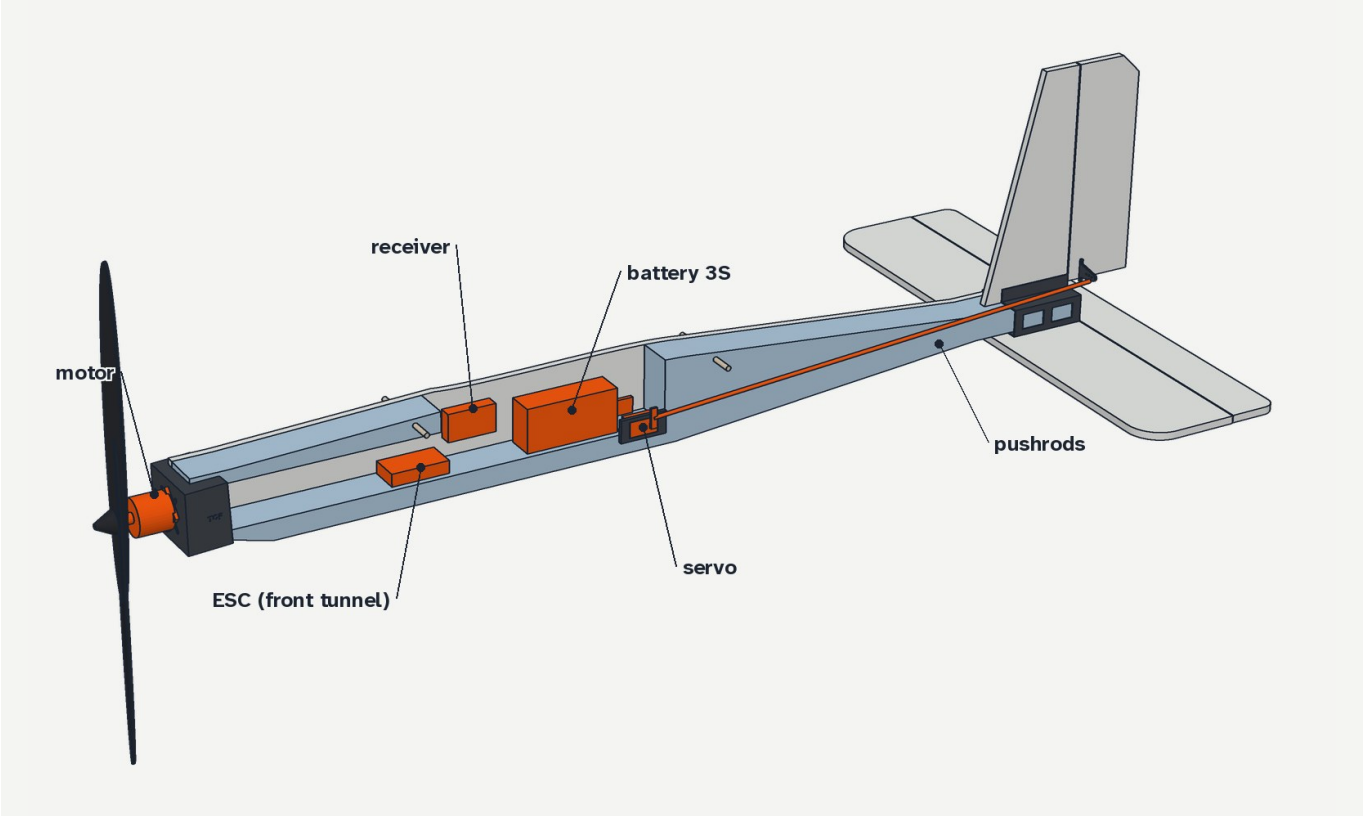
WATCH OUT

No bevel at the hinge. The surfaces cannot move far enough.

Step done ☐

INSTALL THE ELECTRONICS

≈ 1.5 h



YOU NEED ◊3_Wiring diagram · Motor, ESC, 2 servos, receiver · Pushrods, bowden clips, velcro

DO

1. Screw the motor X-mount to the nose part (4 × M3).
2. Put the ESC into the front tunnel. Connect motor and ESC with the 3 bullets.
3. Centre both servos with the transmitter first. Then mount them in the frames and fit the arms.
4. Run the pushrods in their tubes along the sides, fix with the 6 clips. Receiver on velcro.

CHECK ✓

Motor turns freely by hand. Nothing moves when you shake the fuselage.

WATCH OUT

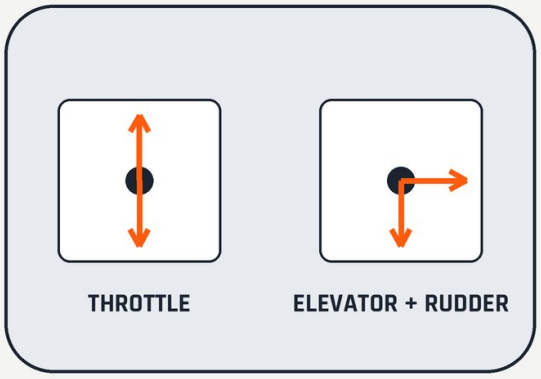
Fitting servo arms before centring. The surfaces end up crooked.

Step done ☐

SET UP THE RADIO

≈ 40 min

Mode 2



Right stick BACK
= elevator UP

Right stick RIGHT
= rudder RIGHT

Left stick UP
= motor faster

Transmitter OFF
= motor stops (failsafe)

Throws (low rate): elevator ± 8 mm, rudder ± 15 mm
Expo 30 %

YOU NEED 04_Firmware_Radio sheet · Transmitter, charged battery · Ruler

DO

1. Bind the receiver. Plug in as the radio sheet shows for your brand.
2. Right stick back = elevator up. Right stick right = rudder right.
3. Set throws: elevator ± 8 mm, rudder ± 15 mm (low rate), 30 % expo.
4. Set failsafe: throttle off. Calibrate the ESC throttle range.

CHECK ✓

Transmitter off: the motor stops within 1 s. All surfaces move the right way.

WATCH OUT

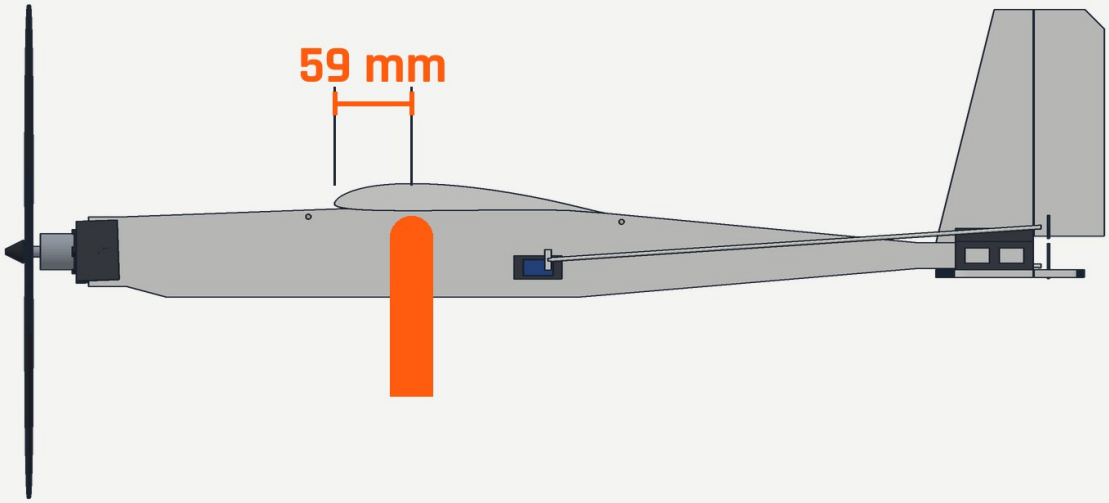
Fixing a reversed surface by turning the horn. Use servo reverse in the transmitter.

Step done ☐

14 BALANCE AND WEIGH

STEP 14 OF 14 · SET UP

≈ 20 min



Hangs level on the fingertips = CG OK (54–67 mm allowed).

YOU NEED Charged 3S battery · Scale · Pen

DO

1. Put on the wing (4 rubber bands straight + 2 crossed).
2. Slide the battery to the mark on the side wall (front edge 260 mm from the nose for 3S 1300).
3. Lift the plane on two fingertips under the wing, 59 mm behind the leading edge.
4. Move the battery until the plane hangs level. Mark the position.

CHECK ✓

Balances level at 59 mm (54–67 mm is OK). Weight 510 g ± 25 g with 3S 1300.

WATCH OUT

Tail-heavy. It flies badly or not at all. When in doubt: a little nose-heavy.

Step done ☐

BEFORE THE FIRST FLIGHT

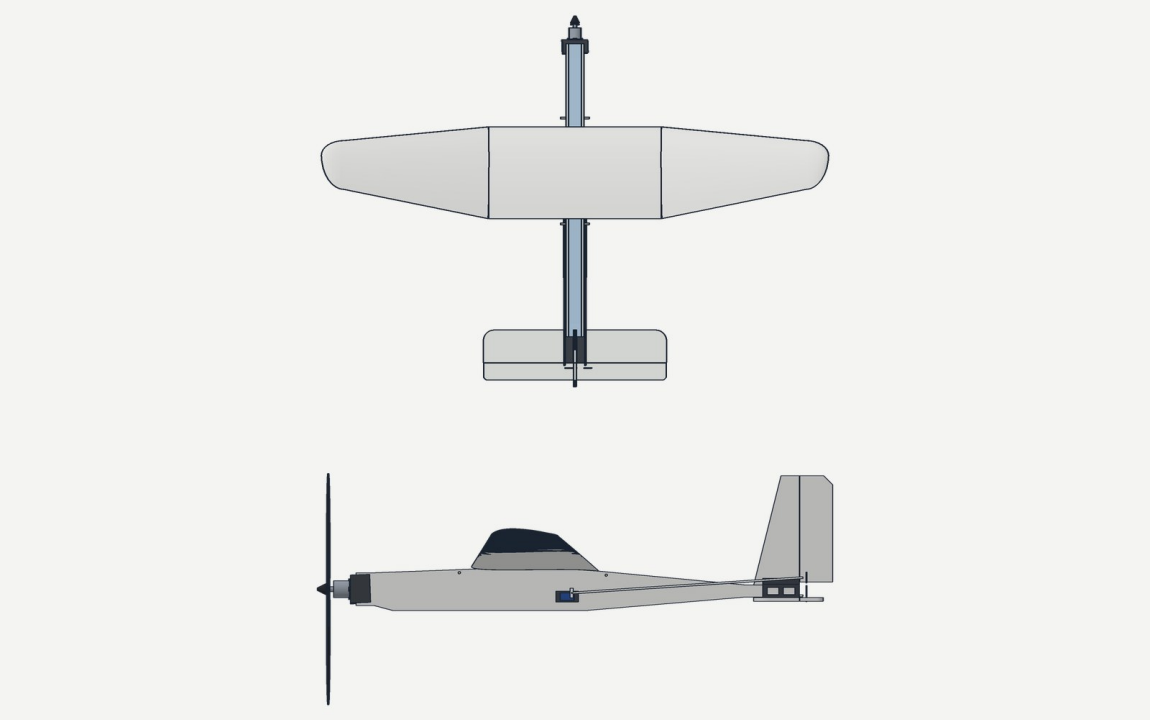
Tick every box. One box open = no flight today.

☐	**RULES**	Registration, insurance and flying site follow your local rules. Fly in sight, away from people.
☐	**BALANCE**	CG at 59 mm behind the leading edge (54–67 mm).
☐	**CONTROLS**	Every surface moves the right way and the right distance. Servo arms and pushrods tight.
☐	**FAILSAFE**	Transmitter off: the motor stops.
☐	**RANGE**	Range check as your radio manual says (reduced power, ≥ 30 m).
☐	**PROPELLER**	Tight, no cracks, pushes air backwards.
☐	**WING**	6 rubber bands, wing centred on the fuselage.
☐	**WEATHER**	Wind under 4 m/s (≈ 15 km/h) for the first flight. Open meadow.
☐	**PILOT**	First battery: an experienced pilot, or after simulator practice. Plan two flight days.
☐	**LAUNCH**	Full throttle, level throw into the wind, wings level. Timer 8 min.

AFTER THE FLIGHT: REPORT BACK

Write down what did not fit, what broke, flight time and battery voltage after landing. Send it as a field report in the catalog. Every report goes into the next revision of this package.

SPECIFICATIONS · REV 1



Wingspan	1160 mm
Wing area	19.86 dm²
Aspect ratio	6.78
Mean aerodynamic chord	181 mm
Airfoil	SD7032 (10 %), 9° dihedral, 2° washout
Fuselage length	730 mm foam + motor
Take-off weight	510 g (3S 1300) · 585 g (3S 2200)
Wing loading	25.7 g/dm²
Balance point (CG)	59 mm behind the leading edge (allowed 54–67 mm)
Controls	3 channels: throttle, elevator, rudder
Motor / prop	2212 ~1000 kV, 3S · 10 × 4.5
Trim speed	34 km/h (calculated)
Flight time	≈ 27 min practice (55 min theory at trim)
Launch / landing	Hand launch · belly landing on grass

Status. Calculated, not yet flown. Rev 1 becomes 'Flight proven' after the first build in the demonstrator workshop. Report anything that does not fit: every report makes the next revision.

Rev 1 · 25 Oct 2026 — First catalog release of the D1b. Printed-rib wing W2 with jig tabs (washout built in), dihedral joiner, tail shoe for 90°, battery marks on the side-wall template.